

$$f(x, y) = x^2 + 4xy + 3y^2 - 6x - 12y$$

to find critical / Stationary points I will find first derivatives with respect to x and y . then I will find OUT WHEN THEY $= 0$

$$f_x(x, y) = 2x + 4y - 6 \quad f_y(x, y) = 4x + 6y - 12$$

$$\begin{array}{l} \text{I. } 2x + 4y - 6 = 0 \\ \text{II. } 4x + 6y - 12 = 0 \quad \times - \frac{1}{2} \\ \hline 2x + 4y - 6 = 0 \\ -2x - 3y + 6 = 0 \\ \hline \end{array} \quad \begin{array}{l} y = 0 \\ \downarrow \text{Substitute } y \text{ in I, and calculate } x \\ 2x + 4(0) - 6 = 0 \\ 2x - 6 = 0 \\ 2x = 6 \\ x = \frac{6}{2} \\ x = 3 \end{array}$$

critical points: $(3, 0)$

to find out if there is: saddle point, local minimum or local maximum I need to do 2nd derivative test

formula is given by book:

$$D = D(x, y) = f_{xx}(x, y) \cdot f_{yy}(x, y) - [f_{xy}(x, y)]^2 \quad \rightarrow \text{this is determinant of Hessian matrix}$$

so I need f_{xx}, f_{xy}, f_{yy} in critical point $(3, 0)$

$$\begin{array}{lll} f_{xx}(x, y) = 2 & f_{xy}(x, y) = 4 & f_{yy}(x, y) = 6 \\ f_{xx}(3, 0) = 2 & f_{xy}(3, 0) = 4 & f_{yy}(3, 0) = 6 \end{array}$$

$$D(3, 0) = f_{xx}(3, 0) \cdot f_{yy}(3, 0) - [f_{xy}(3, 0)]^2$$

$$D(3, 0) = 2 \cdot 6 - (4)^2$$

$$D(3, 0) = 12 - 16$$

$$D(3, 0) = -4$$

by the rules: $D > 0$ and $f_{xx}(a, b) > 0$ then $f(a, b) = \text{Local MIN}$
 $D > 0$ and $f_{xx}(a, b) < 0$ then $f(a, b) = \text{Local Max}$
 $D < 0$ then $(a, b) = \text{saddle point of } f$

$$\downarrow \quad \Rightarrow \quad D(3, 0) = -4 \Rightarrow \boxed{D(3, 0) < 0} \Rightarrow \underline{\text{saddle point}}$$

$$V = x \cdot y \cdot z$$

$x \rightarrow \text{length}$

$y \rightarrow \text{width}$

$z \rightarrow \text{height}$

$$x + y + z = 30 \text{ cm}$$

a) $V = x \cdot y \cdot z \wedge x + y + z = 30 \text{ cm}$

so I think max volume will be if $x = y = z = 10 \text{ cm}$ (because I will never get higher than 1000)

so $V_{\max} = 10 \cdot 10 \cdot 10 = 1000 \text{ cm}^3$

b) Lagrange

Lagrange: $\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \rightarrow$ (Gradient vector of f is parallel with grad. vector of g) (if $\nabla g \neq 0$)



Gradient: $\nabla f(x, y, z) = (f_x(x, y, z), f_y(x, y, z), f_z(x, y, z))$

$$f(x, y, z) = x \cdot y \cdot z$$

$$g(x, y, z) = x + y + z = 30 \text{ cm}$$

$$\downarrow \quad f_x(x, y, z) = y \cdot z \quad f_y(x, y, z) = x \cdot z \quad f_z(x, y, z) = x \cdot y \quad \left| \quad g_x(x, y, z) = 1 \quad g_y(x, y, z) = 1 \quad g_z(x, y, z) = 1 \right.$$



$$\nabla f(x, y, z) = (y \cdot z, x \cdot z, x \cdot y)$$

$$\nabla g(x, y, z) = (1, 1, 1)$$

$$f_x(x, y, z) = \lambda g_x(x, y, z)$$

$$f_y(x, y, z) = \lambda g_y(x, y, z)$$

$$f_z(x, y, z) = \lambda g_z(x, y, z)$$

$$\downarrow$$

$yz = 2 \cdot 1$
$xz = 2 \cdot 1$
$xy = 2 \cdot 1$

$$\rightarrow yz = xz = xy = 2$$

$$\swarrow$$

$$x = y = z$$

$$x + y + z = 30 \text{ cm}$$

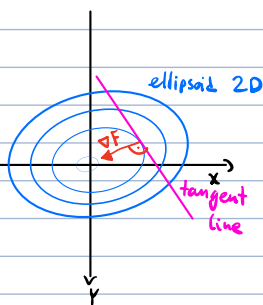
$$x + x + x = 30 \text{ cm}$$

$$3x = 30 \text{ cm}$$

$$x = \frac{30}{3} \text{ cm}$$

$$\boxed{x = 10 \text{ cm}} \text{ Since } x = y = z \Rightarrow \text{max Volume is } 10 \cdot 10 \cdot 10 = 1000 \text{ cm}^3$$

$$F(x, y, z) = a(x-d)^2 + b(y-e)^2 + c(z-f)^2 = L$$



a) gradient vector $\rightarrow$ this normal vector is perpendicular to tangent line in specific point (Plane)

c) Now I will use formula for 3D space $a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$

$\rightarrow a, b, c \rightarrow$ normal vector to the plane (I will use ∇F)

$\rightarrow x, y, z \rightarrow$ any point in 3D space

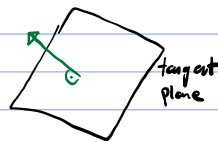
$\rightarrow x_0, y_0, z_0 \rightarrow$ Point on Plane (here I will put point that is on ellipsoid to get: tangent plane)

$$a) \nabla F = (F_x(x, y, z), F_y(x, y, z), F_z(x, y, z))$$

$$F_x = 2a(x-d) \quad F_y = 2b(y-e) \quad F_z = 2c(z-f)$$

$$\rightarrow \text{so } \nabla F = (2a(x-d), 2b(y-e), 2c(z-f))$$

$\downarrow$ now evaluate the gradient vector at point $(x_0, y_0, z_0) \rightarrow$ to get normal vector to tangent plane at point



$$\nabla F(x_0, y_0, z_0) = (2a(x_0-d), 2b(y_0-e), 2c(z_0-f))$$

$\downarrow$ by using this normal vector, to the tangent plane at some point (x_0, y_0, z_0) I can plug it in plane eq:

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$\downarrow$

$$2a(x_0-d) \cdot (x-x_0) + 2b(y_0-e) \cdot (y-y_0) + 2c(z_0-f) \cdot (z-z_0) = 0 \quad \checkmark \cdot 1/2$$

$$a(x_0-d)(x-x_0) + b(y_0-e)(y-y_0) + c(z_0-f)(z-z_0) = 0$$

$\downarrow$ Coef. $\rightarrow$ Shape of ellipse
 $\downarrow$ Center of ellipse
 $\downarrow$ any point on tangent plane
 $\downarrow$ Point on the ellipsoid where tangent plane touches the ellipsoid